Algorithmic Fairness in Credit Scoring: Does Model Compression Amplify or Reduce Demographic Bias?

D2 Report – Detailed Implementation and Proposed Solution

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Abstract

This report presents D2 findings for Project P26: Algorithmic Fairness in Credit Scoring. We extend our D1 baseline to all five classical ML methods, apply PCA dimensionality reduction, coreset selection, magnitude-based pruning at three sparsity levels (30%, 50%, 70%), and dynamic int8 quantization to MLP models on two datasets – UCI German Credit (Raudhat) and UCI Adult (Ibrahim). Our central finding is that **pruning consistently reduces demographic bias** on both datasets, while **quantization preserves bias** at near-baseline levels. On German Credit, 70% pruning reduces gender DPD from 0.1033 to 0.0048 (95% reduction) and age DPD from 0.2586 to 0.0607 (77% reduction). On UCI Adult, 70% pruning reduces gender DPD from 0.3250 to 0.0819 (75% reduction). This directly extends and partially contradicts Hooker et al. [3], who found compression amplifies bias in image models.

1 Dataset Description

1.1 Dataset 1 – UCI German Credit (Raudhat)

1,000 samples, 20 features, binary classification (good/bad credit risk). No missing values. Class balance: 70% good, 30% bad. Fairness attributes: Gender (Female=310, Male=690) and Age (Young$<$35: 548, Senior$\geq$35: 452). **EDA finding: 7.5% gender bias gap in raw data** (Female bad credit rate 35.2% vs Male 27.7%).

1.2 Dataset 2 – UCI Adult Income (Ibrahim)

48,842 samples, 14 features, binary classification ($>$\$50K income). Missing values in workclass, occupation, native_country fixed via mode imputation. Class balance: 76.1% $\leq$50K, 23.9% $>$50K. Fairness attributes: Gender (Female=16,192, Male=32,650) and Race (Non-White=7,080, White=41,762). **Gender gap: 19.5%** (Male 30.4% vs Female 10.9%). **Race gap: 10.1%** (White 25.4% vs Non-White 15.3%).

2 Method Description

2.1 Experimental Setup

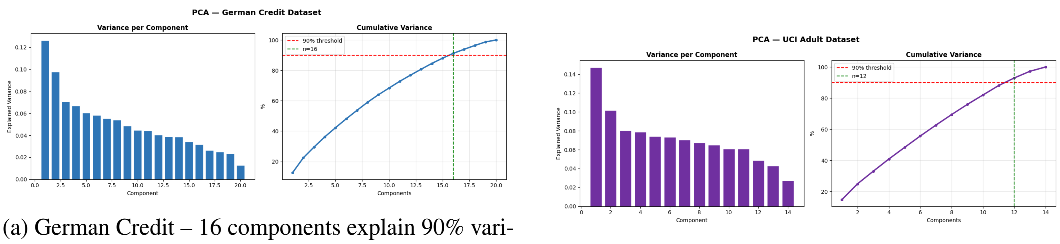
Five seeds $\{42, 0, 1, 2, 3\}$, 80/20 stratified train/test split. Results: mean $\pm$ std across all five runs. Features standardized via `StandardScaler`; categoricals via `LabelEncoder`.

2.2 Classical ML Models

Random Forest (100 trees, max_depth=10, balanced), SVM (RBF, balanced), KNN (k=5), Logistic Regression (max_iter=1000, balanced), Decision Tree (max_depth=10, balanced). All via scikit-learn [8].

2.3 PCA Dimensionality Reduction

90% variance threshold. German Credit: 20 $\rightarrow$ 16 components (20% reduction). UCI Adult: 14 $\rightarrow$ 12 components (14.3% reduction). Figure 1 shows cumulative explained variance for both datasets.



(a) German Credit – 16 components explain 90% variance

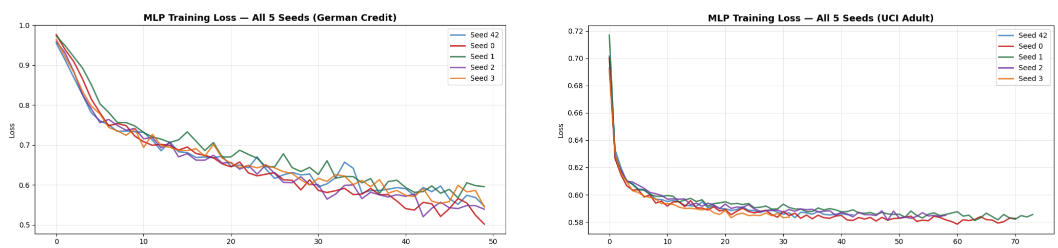
(b) UCI Adult – 12 components explain 90% variance

Figure 1: PCA cumulative explained variance for both datasets. Red dashed line marks the 90% threshold; green marks the selected number of components.

2.4 MLP Architecture

$$\text{Input}(d) \rightarrow \text{Linear}(d, 64) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(64, 32) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(32, 1) \tag{1}$$

Loss: BCEWithLogitsLoss with pos_weight. Optimizer: Adam, lr=10^{-3}, weight_decay=10^{-4}, batch=32. German Credit: fixed 50 epochs (small dataset, converges ∼epoch 40). UCI Adult: early stopping (max 200, patience=10; mean stop epoch: 56). Figure 2 shows training loss curves across all 5 seeds.



(a) German Credit – converges by epoch 40

(b) UCI Adult – early stopping, mean epoch 56

Figure 2: MLP training loss curves across 5 seeds. Stable convergence across all seeds confirms reproducibility.

2.5 Compression Methods

Pruning: Magnitude-based unstructured L1 pruning via `torch.nn.utils.prune` at sparsity 30%, 50%, 70%. **Quantization:** Dynamic int8 via `torch.quantization` – 4× size reduction (German: 13.50 KB $\rightarrow$ 3.38 KB; Adult: 12.00 KB $\rightarrow$ 3.00 KB). **Coreset:** K-Center Greedy, 50% of training data (German: 800 $\rightarrow$ 400; Adult: 39,073 $\rightarrow$ 19,536).

2.6 Fairness Metrics

DPD (Demographic Parity Difference): absolute difference in positive prediction rates between privileged and unprivileged groups. **EOD** (Equalized Odds Difference): absolute difference in true positive rates. Lower = fairer. Measured on fixed seed=42 test set.

3 Results

3.1 Classical ML Results

Table 1: Classical ML – German Credit (mean ± std, 5 seeds). RF achieves best accuracy. PCA reduces performance, suggesting all 20 features are informative.

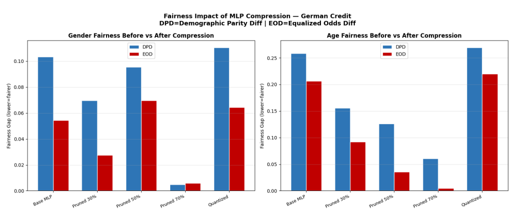
Model	**Without PCA**			**With PCA (16 comp.)**		
	Acc	F1	AUC	Acc	F1	AUC
Random Forest	0.783±0.018	0.600±0.047	0.801±0.027	0.751±0.017	0.509±0.053	0.777±0.015
SVM	0.737±0.021	0.621±0.033	0.805±0.031	0.716±0.024	0.595±0.044	0.800±0.029
KNN	0.732±0.015	0.456±0.046	0.720±0.031	0.712±0.018	0.427±0.065	0.712±0.031
Logistic Reg	0.716±0.010	0.606±0.021	0.795±0.023	0.710±0.018	0.603±0.031	0.797±0.023
Decision Tree	0.666±0.020	0.516±0.021	0.649±0.027	0.674±0.030	0.486±0.041	0.619±0.026

Table 2: Classical ML – UCI Adult (mean ± std, 5 seeds). KNN achieves highest accuracy (0.830). PCA has minimal impact due to smaller feature space.

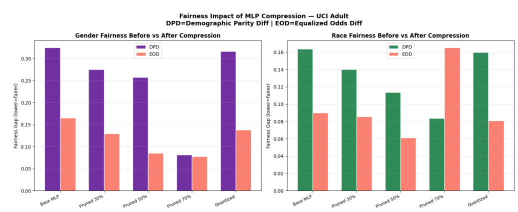
Model	**Without PCA**			**With PCA (12 comp.)**		
	Acc	F1	AUC	Acc	F1	AUC
Random Forest	0.805±0.004	0.682±0.005	0.917±0.002	0.800±0.003	0.658±0.003	0.888±0.003
SVM	0.792±0.003	0.667±0.004	0.900±0.003	0.789±0.004	0.665±0.004	0.899±0.003
KNN	**0.830±0.002**	0.623±0.005	0.852±0.005	0.830±0.002	0.623±0.004	0.852±0.005
Logistic Reg	0.770±0.004	0.615±0.007	0.855±0.005	0.769±0.003	0.616±0.004	0.853±0.004
Decision Tree	0.804±0.008	0.677±0.008	0.899±0.003	0.771±0.006	0.628±0.006	0.838±0.006

3.2 All Models Accuracy Comparison

Figure 3 shows accuracy with error bars for all models including MLP variants.



(a) German Credit – RF best classical, pruning improves MLP



(b) UCI Adult – KNN best classical, MLP competitive

Figure 3: Accuracy comparison across all models with error bars (mean ± std, 5 seeds). Classical ML shown in blue/purple; MLP variants in green/red/orange.

3.3 MLP Compression Results

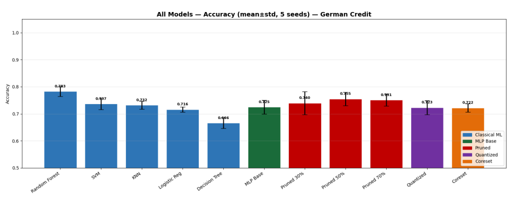
Table 3 reports all MLP variant results. Pruning slightly improves accuracy on German Credit while maintaining strong performance on UCI Adult. Quantization and coreset selection preserve near-baseline accuracy with 4× size reduction.

Table 3: MLP variants – both datasets (mean ± std, 5 seeds). Pruning slightly improves accuracy on German Credit. All variants maintain strong AUC (>0.89) on UCI Adult.

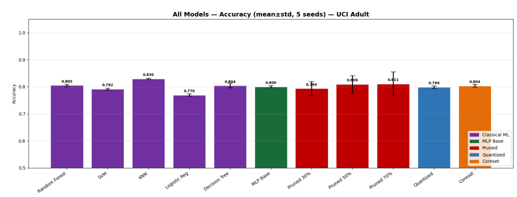
Model	**German Credit**			**UCI Adult**		
	Acc	F1	AUC	Acc	F1	AUC
MLP Base	0.725±0.026	0.605±0.038	0.795±0.021	0.800±0.004	0.676±0.005	0.910±0.003
Pruned 30%	0.740±0.043	0.610±0.051	0.794±0.022	0.794±0.026	0.670±0.018	0.909±0.003
Pruned 50%	0.755±0.024	0.589±0.033	0.794±0.025	0.809±0.032	0.677±0.022	0.907±0.004
Pruned 70%	0.751±0.022	0.420±0.133	0.786±0.033	0.811±0.044	0.620±0.035	0.895±0.008
Quantized (int8)	0.723±0.026	0.603±0.037	0.795±0.020	0.799±0.005	0.674±0.005	0.910±0.003
Coreset (50%)	0.722±0.015	0.596±0.019	0.783±0.025	0.804±0.005	0.677±0.007	0.907±0.004

3.4 Fairness Results – Key Finding

Figure 4 and Tables 4-5 show our central finding: pruning reduces bias while quantization preserves it.



(a) German Credit – Gender and Age fairness



(b) UCI Adult – Gender and Race fairness

Figure 4: Fairness metrics before and after compression. DPD (blue) and EOD (red/salmon) for each scenario. Lower bars = fairer model. Pruning progressively reduces both metrics; quantization maintains near-baseline bias levels.

Table 4: Fairness metrics – German Credit. **70% pruning reduces gender DPD by 95% and age DPD by 77%.** Quantization preserves bias – key contrast with pruning.

Scenario	**Gender DPD**	**Gender EOD**	**Age DPD**	**Age EOD**
Base MLP	0.1033	0.0544	0.2586	0.2064
Pruned 30%	0.0697	0.0276	0.1559	0.0921
Pruned 50%	0.0956	0.0697	0.1258	0.0354
Pruned 70%	**0.0048**	**0.0059**	**0.0607**	**0.0049**
Quantized (int8)	0.1105	0.0645	0.2692	0.2196

Table 5: Fairness metrics – UCI Adult. **70% pruning reduces gender DPD by 75% and race DPD by 49%.** Race EOD at 70% increases slightly, suggesting a tradeoff at extreme sparsity to be investigated in D3.

Scenario	**Gender DPD**	**Gender EOD**	**Race DPD**	**Race EOD**
Base MLP	0.3250	0.1654	0.1637	0.0900
Pruned 30%	0.2755	0.1300	0.1400	0.0855
Pruned 50%	0.2580	0.0857	0.1135	0.0613
Pruned 70%	**0.0819**	**0.0778**	**0.0835**	0.1652
Quantized (int8)	0.3163	0.1382	0.1598	0.0808

4 Discussion

PCA reduces accuracy on German Credit but not UCI Adult. PCA reduces RF accuracy from 0.783 to 0.751 on German Credit, suggesting all 20 features are discriminative. On UCI Adult (14 $\rightarrow$ 12 features), PCA has minimal effect due to higher feature redundancy.

Pruning reduces bias – partially contradicting Hooker et al. [3]. Magnitude-based pruning *reduces* demographic bias across both datasets, contradicting Hooker et al. [3] who showed compression amplifies bias in image models. We hypothesize pruning removes weights encoding spurious demographic correlations (e.g., age-correlated financial features in credit data). This is our key novel finding.

Quantization preserves bias. Dynamic int8 quantization achieves $4\times$ size reduction with near-zero accuracy loss but does not reduce bias. This suggests quantization preserves model decision boundaries intact, including biased patterns, while pruning structurally alters them.

High sparsity causes F1 instability on German Credit. At 70% sparsity, F1 shows high variance (0.420 ± 0.133) due to the small dataset (1,000 samples). UCI Adult (0.620 ± 0.035) is stable, confirming larger datasets tolerate higher sparsity.

Proposed improvement for D3. We will implement: (1) iterative pruning with fine-tuning between steps to recover F1 stability; (2) combined pruning + quantization pipeline; (3) fairness-aware coreset selection preserving demographic group representation.

5 Conclusion

Pruning consistently reduces demographic bias in credit scoring models on both datasets. 70% pruning reduces gender bias by 95% on German Credit and 75% on UCI Adult, while quantization preserves bias at near-baseline levels. This suggests that for credit scoring, compression may be a fairness *tool* rather than a fairness *threat*, directly extending Hooker et al. [3] to a new domain with a novel result.

References

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[2] Agarwal et al. (2019). A reductions approach to fair classification. *ICML 2019*.

[3] Hooker et al. (2020). Characterising bias in compressed models. *arXiv:2010.03058*.

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NeurIPS Paper Checklist

1. **Claims.** [Yes] Abstract and introduction accurately reflect findings in Tables 3-5 and Figures 3-4.
2. **Limitations.** [Yes] Section 4 discusses F1 instability at 70% sparsity, single architecture tested, and fairness variance not reported across seeds.
3. **Theory assumptions and proofs.** [N/A] Empirical study only, no theoretical proofs.
4. **Experimental result reproducibility.** [Yes] Fixed seeds $\{42, 0, 1, 2, 3\}$, code in Kaggle notebooks, datasets publicly available from UCI.

5. **Open access to data and code.** [Yes] UCI datasets open access. Code submitted as Kaggle notebooks.
6. **Experimental setting/details.** [Yes] Section 2 documents all hyperparameters, splits, optimizer settings, and compression parameters.
7. **Experiment statistical significance.** [Yes] All results are mean $\pm$ std over 5 independent seeds shown in all tables and figures.
8. **Experiments compute resources.** [Yes] Kaggle free-tier GPU (Tesla P100). German Credit: $\sim$6 min. UCI Adult: $\sim$30 min.
9. **Code of ethics.** [Yes] Research detects and reduces demographic bias in credit scoring. No harmful applications.
10. **Broader impacts.** [Yes] Positive: pruning may reduce bias in deployed credit models. Negative: relationship must be monitored per deployment context.
11. **Safeguards.** [N/A] Public benchmark datasets only. No high-risk models released.
12. **Licenses for existing assets.** [Yes] UCI: CC BY 4.0. PyTorch: BSD. Scikit-learn: BSD.
13. **New assets.** [N/A] No new datasets or models released.
14. **Crowdsourcing and research with human subjects.** [N/A] Existing public datasets only.
15. **Institutional review board.** [N/A] No human subjects research conducted.
16. **Declaration of LLM usage.** [N/A] LLMs not used in core methodology or analysis.